

ASSIGNMENT -1

[1] ✓ 23s #1. Find area of rectangle

```
length = float(input("Enter length: "))
width = float(input("Enter width: "))

area = length * width

print("Area of rectangle =", area)
```

▼ Enter length: 20
Enter width: 10
Area of rectangle = 200.0

[2] ✓ 13s #2. Find simple interest

```
principal = float(input("Enter principal amount: "))
rate = float(input("Enter rate of interest: "))
time = float(input("Enter time (in years): "))

simple_interest = (principal * rate * time) / 100

print("Simple Interest =", simple_interest)
```

▼ ... Enter principal amount: 200
Enter rate of interest: 30
Enter time (in years): 20
Simple Interest = 1200.0

[3] ✓ 5s #3. Convert Celsius to Fahrenheit

```
celsius = float(input("Enter temperature in Celsius: "))

fahrenheit = (celsius * 9/5) + 32

print("Temperature in Fahrenheit =", fahrenheit)
```

▼ Enter temperature in Celsius: 20
Temperature in Fahrenheit = 68.0

[4] ✓ 17s #4. Calculate average of 3 numbers

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
num3 = float(input("Enter third number: "))

average = (num1 + num2 + num3) / 3

print("Average =", average)
```

▼ Enter first number: 44
Enter second number: 34
Enter third number: 24
Average = 34.0

[5] ✓ 3s #5. Find square and cube of a number

```
number = float(input("Enter a number: "))

square = number ** 2
cube = number ** 3

print("Square =", square)
print("Cube =", cube)
```

▼ Enter a number: 5
Square = 25.0
Cube = 125.0

[6] ✓ 7s #6. Swap two numbers without using third variable

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("Before swapping:")
print("a =", a)
print("b =", b)

a, b = b, a

print("After swapping:")
print("a =", a)
print("b =", b)
```

```
[6] ✓ 7s
print("After swapping:")
print("a =", a)
print("b =", b)
```

Enter first number: 50
Enter second number: 40
Before swapping:
a = 50
b = 40
After swapping:
a = 40
b = 50

```
[7] ✓ 21s #7. Student Report Program
```

```
# Taking student details
student_name = input("Enter student name: ")
roll_number = input("Enter roll number: ")

# Taking marks of 3 subjects
mark1 = float(input("Enter marks in Subject 1: "))
mark2 = float(input("Enter marks in Subject 2: "))
mark3 = float(input("Enter marks in Subject 3: "))

# Calculating total marks
total = mark1 + mark2 + mark3

# Calculating percentage
percentage = (total / 300) * 100
```

```
[7] ✓ 21s # Calculating total marks
```

```
total = mark1 + mark2 + mark3
```

```
# calculating percentage
percentage = (total / 300) * 100
```

```
# Displaying report
print("\n----- STUDENT REPORT -----")
print("Student Name :", student_name)
print("Roll Number :", roll_number)
print("Subject 1 Marks :", mark1)
print("Subject 2 Marks :", mark2)
print("Subject 3 Marks :", mark3)
print("Total Marks :", total)
print("Percentage :", percentage, "%")
```

... Enter student name: Avni
Enter roll number: 60
Enter marks in Subject 1: 89
Enter marks in Subject 2: 97
Enter marks in Subject 3: 95

```
----- STUDENT REPORT -----
Student Name : Avni
Roll Number : 60
Subject 1 Marks : 89.0
Subject 2 Marks : 97.0
Subject 3 Marks : 95.0
Total Marks : 281.0
Percentage : 93.66666666666667 %
```